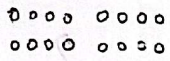
# States of matter and Atomic structure

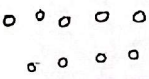
① Solid
② liquid
③ Gas
④ plasma

## Crystaline Solid



① High mp, bp
② Strong
③ rigid
④ Ex: NaCl

## Amorphos solid



① low mp, bp
② weak
③ brittle
④ Ex: Glass

## Mole

1 Mole = $6.022 \times 10^{23}$ atoms / molecule / ions

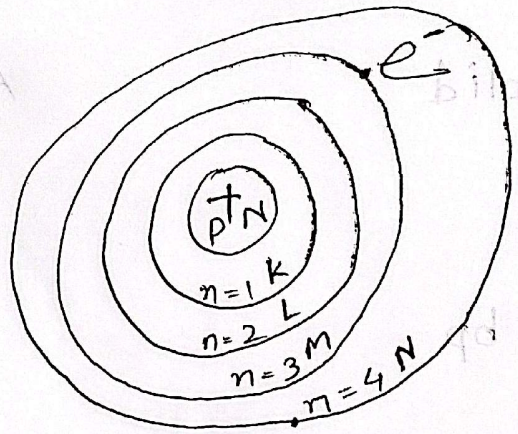
= Molecular Mass / Atomic Mass

= 22.4 L for gas at STP.

O = 16g = 1 mole atom

$O_2$ = 32g = 1 mole molecule

## Atomic Structure




2
8
18
32

## Quantum Number

① principle Q.N (n)

→ define position and energy of e⁻

→ Allowed values, n = 1, 2, 3, 4....

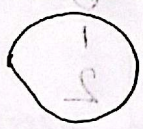
→ electron - $2n^2$

(II) Azimethal Q.N (l)

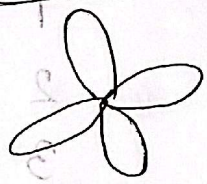
→ Shape of orbitals

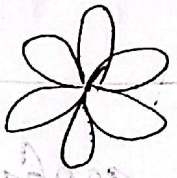
→ Allowed values, l = 0 .... (n-1)

→ electron 2(2l+1)

l = 0 ; s orbital ,  Spherical

l = 1, p orbital  dumble

l = 2 d orbital  double dumble

l = 3 f orbital  complex shap

(III) Magnetic Q.N (m)

→ define orientation of orbital

→ -l ---- 0 --- +l

(IV) Spin Q.N (s)

→ spin direction of e-

→ $s = \pm \frac{1}{2}$

Class-2

| n | L | m | orbital |
|---|---|---|---|
| 1 | 0 | 0 | 1S |
| 2 | 0 | 0 | 2S |
| | 1 | -1, 0, +1 | 2P ($P_x, P_y, P_z$) |
| 3 | 0 | 0 | 3S |
| | 1 | -1, 0, +1 | 3P |
| | 2 | -2, -1, 0, +1, +2 | 3d ($d_{xy}, d_{yz}, d_{zx}, d_{x^2-y^2}, d_{z^2}$) |
| 4 | 0 | 0 | 4S |
| | 1 | -1, 0, +1 | 4P |
| | 2 | -2, -1, 0, +1, +2 | 4d |
| | 3 | -3, -2, -1, 0, +1, +2, +3 | 4f |

Electron configuration

$K_{19} \rightarrow 1S^2\ 2S^2\ 2P^6\ 3S^2\ 3P^6\ 4S^1$ → $3d^0\ 4S^1$

n+1 ক্ব value যার কম সেটা, অর্থাৎ e⁻ যাবে

$Cr(24) \rightarrow 1S^2\ 2S^2\ 2P^6\ 3S^2\ 3P^6\ 3d^5\ 4S^1$ : Half filled, Full filled, Stable তাকে

1S 2S 2P 3S 3P 4S 3d

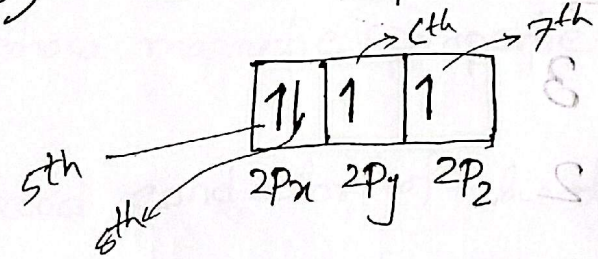
4P 5S 4d 5P 6S

Nb (41) → $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 3d^{10}\ 4s^2\ 4p^6\ 4d^3\ 5s^2$

[↑↓ | ↑ | ] but [$4d^4\ 5s^1$] more stable

## Hunds Rule [সর্বোচ্চ বিজোড় সংখ্যক থাকবে]

O(8) – $1s^2\ 2s^2\ 2p^4$




**5th e⁻**

$n = 2$

$l = 1$

$m = -1$

$s = +\frac{1}{2}$

**8th e⁻**

$n = 2$

$l = 1$

$m = -1$

$s = -\frac{1}{2}$

## Paulis exclusion law [2টি e⁻ এর চারটি কোয়ান্টাম সংখ্যা সমান হওয়া কোনভাবেই হবে না]

# Periodic Table

## Properties of d-block elements (Gr 3 - Gr 12)

(i) Show variable oxidation state due to vaccant d-orbit

(ii) Forms colored compount with ligands due to d-d transition

ligands → can donate e⁻ pair.

(iii) Act as catalyst.

(iv) Show magnetic Properties. (paramagnetism & Diamagnetism)

(v) Good conductor of heat and electricity

(vi) Forms Alloys.

## # Period & Group Determination

$Fe(26) \rightarrow [Ar]\ 4s^2\ 3d^6$

period = height principle Q.N.

= 4

Group

S-block (Gr 1 / Gr 2)

$Mg(12) - 1s^2\ 2s^2\ 2p^6\ 3s^2$

↳ Gr = 2, P = 3

P-block

. Gr = 10 + valence $e^-$

O (8) – $1s^2 2s^2 2p^4$

Gr = 16

d-block

Gr = n s + (n-1) d

Fe (26) – [Ar] $4s^2 3d^6$

Gr = 4s + 3d

= 2 + 6

= 8

As (33) – [Ar] $3d^{10} 4s^2 4p^3$

Gr = 15

* properties & uses of nobel Gas